

Lanezy — Future Scope

1. Real-world deployment & scaling

- Pilot rollouts with municipal traffic departments (1–3 intersections → district → city).
- Edge-first architecture: deploy compact YOLO inference on edge devices to reduce bandwidth and maintain privacy.
- Horizontal scaling to support hundreds → thousands of intersections with distributed controllers and centralized analytics.

2. Integration with civic infrastructure

- Integrate with traffic controllers, city GIS, emergency dispatch, and municipal work-order systems for pothole repair.
- Connect with ambulance dispatch, ride-hailing, mapping providers, and tow networks via third-party APIs.
- Provide anonymized feeds to city planners for infrastructure decisions.

3. Technical enhancements & research directions

- Adaptive hybrid control: combine PRR-MASC with lightweight RL/A2C fallback for complex intersections.
- Multi-camera fusion & tracking to improve occlusion handling and turn-probability estimation.
- Predictive maintenance using historical YOLO density and pothole reports to forecast road degradation.

4. New modules & product expansion

- Predictive congestion alerts for commuters and apps.
- Dynamic routing and prioritization for public transport (buses).
- Automated enforcement workflows: human-in-the-loop verification → challan/towing dispatch.

5. Data & analytics platform

- Role-based dashboards (city planners, traffic officers, public transparency).
- KPI library: average wait time, throughput, starvation incidents, override success rate, pothole-resolve time, false-positive rates.
- Anomaly detection to auto-trigger incident workflows for unusual traffic patterns.

6. Field testing, evaluation & validation

- A/B trials comparing PRR-MASC vs baseline timings and commercial adaptive controllers.
- Safety validation: failure-mode testing and actuator safety checks for barrier systems.
- User studies for app features (tow service effectiveness, chatbot accuracy, multilingual usability).

7. Legal, regulatory & ethical considerations

- Privacy-first design: on-edge anonymization, store only necessary metadata, clear retention policies.
- Compliance with local laws for video surveillance, challan issuance, and data sharing.
- Transparent auditing for overrides, barrier operations, and enforcement decisions.

8. Commercialisation & sustainability

- Business models: SaaS for cities, hardware-as-a-service, premium analytics packages, and government grants.
- Partnerships with traffic equipment vendors, municipal corporations, ambulance services, and telecoms.
- Revenue streams: licensing, data services to insurers/logistics, paid tow & analytics features.

9. Social impact & long-term goals

- Reduced fatalities & faster emergency response through green corridors and enforcement.
- Evidence-driven pothole repair priorities and measurable ROI for roadworks.
- Inclusive features: multilingual chatbot (Hindi/Odia/English) to increase accessibility.

10. Academic & R&D; outputs

- Publish PRR-MASC results, hybrid-control comparisons, and pothole-sensing studies.
- Share carefully anonymized datasets for the research community (privacy-compliant).
- Partner with universities for CV, sensor-fusion, and urban-computing research.

Recommended near-term roadmap

Timeline	Focus
3 months	Pilot 2–4 intersections; tow/chatbot live beta; collect baseline KPIs.
6 months	Full pilot evaluation; safety audit for barrier/override; improve OCR and multilingual chatbot.
12 months	Expand pilot city-wide; integrate with municipal systems; start commercial conversations.

Key success metrics to track

- Mean intersection delay (s)
- Throughput (vehicles/hour) per lane
- Emergency override time saved (s) / number of successful clearances
- Pothole report to resolve time (days)
- False-positive rate for violations (%)
- User satisfaction (app & chatbot) — CSAT